Examiner only

**2.** In 1992 scientists first detected a planet orbiting a star outside our Solar System. Hundreds of these exoplanets have been found since. **Table 1** shows the stars that have been discovered with 6 or more exoplanets in orbit, along with information about our Sun.

**Table 1**

| Name of star | Distance of star from Earth (l-y) | Mass of star (compared to the Sun) | Temperature of star (K) | Number of planets in orbit |
|---|---|---|---|---|
| Sun | 0.00002 | 1.0 | 5770 | 8 |
| HR 8832 | 21 | 0.8 | 4700 | 7 |
| HD 10180 | 127 | 1.1 | 5910 | 7 |
| Kepler 90 | 2500 | 1.1 | 5930 | 7 |
| HD 40307 | 42 | 752.0 | 4980 | 6 |
| HD 127 | 206 | 0.7 | 870 | 6 |
| Kepler 20 | 950 | 0.9 | 5470 | 6 |
| Kepler 11 | 2000 | 1.0 | 5680 | 6 |

*(a)* (i) Tick (✓) the **two** correct statements below. [2]

Four stars have the same mass. ☐

Light travelling from Kepler 90 takes the longest time to reach Earth. ☐

The range of star temperatures is 5060 K. ☐

Hotter stars have more planets in orbit around them. ☐

Star HR 8832 is double the distance of star HD 40307 away from Earth. ☐

(ii) State **two** reasons, using evidence from **Table 1**, why scientists predict that the star Kepler 11 is most likely to have a Solar System similar to ours. [2]

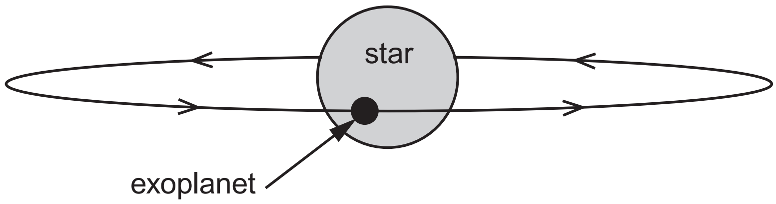
..............................................................................................................................................

..............................................................................................................................................

..............................................................................................................................................

Examiner only

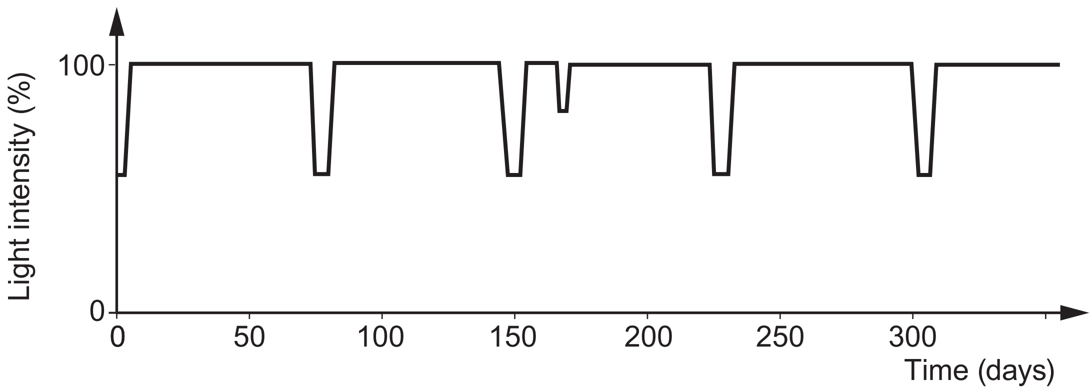
*(b)* Exoplanets can't be detected directly because they don't give out their own light. One method of detection is to look for a slight dip in the light intensity from the star. This happens when the exoplanet passes in front of the star during its orbit. This is called a transit.




(i) Why does the detected light intensity decrease during a transit? [1]

..........................................................................................................................................

(ii) A pupil finds a graph on the internet that shows the measured light intensity detected from a star against time.




I. Calculate the orbit time, in days, of the exoplanet. [2]

Orbit time = .............................. days

II. The pupil correctly suggests that there may be two exoplanets in orbit around this star. What evidence is there to back up this claim? [1]

..........................................................................................................................................

III. During a transit some of the light from the star passes through the exoplanet's atmosphere. An absorption spectrum is produced. Explain how the absorption spectrum arises and is used to identify gases in its atmosphere. [2]

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

3420UB01
07

Examiner only

*(c)* **Table 2** shows data collected for all the exoplanets in orbit around the star called Kepler 11. They are listed in order of increasing distance.

**Table 2**

| Exoplanet | Mass (compared to Earth) | Time for one orbit (days) | Temperature (K) | Orbit radius (AU) |
|---|---|---|---|---|
| b | 1.9 | 10.3 | 871 | 0.09 |
| c | 2.9 | 13.0 | 807 | 0.11 |
| d | 7.3 | 22.7 | 659 | 0.16 |
| e | 8.0 | 32.0 | 596 | 0.20 |
| f | 2.0 | 46.7 | 525 | .............................. |
| g | 25.0 | 118.4 | 386 | 0.47 |

(i) Compare the **trend** in temperature with orbit radius of the planets around Kepler 11 to the planets in our Solar System where Venus is the hottest. [2]

.............................................................................................................................................

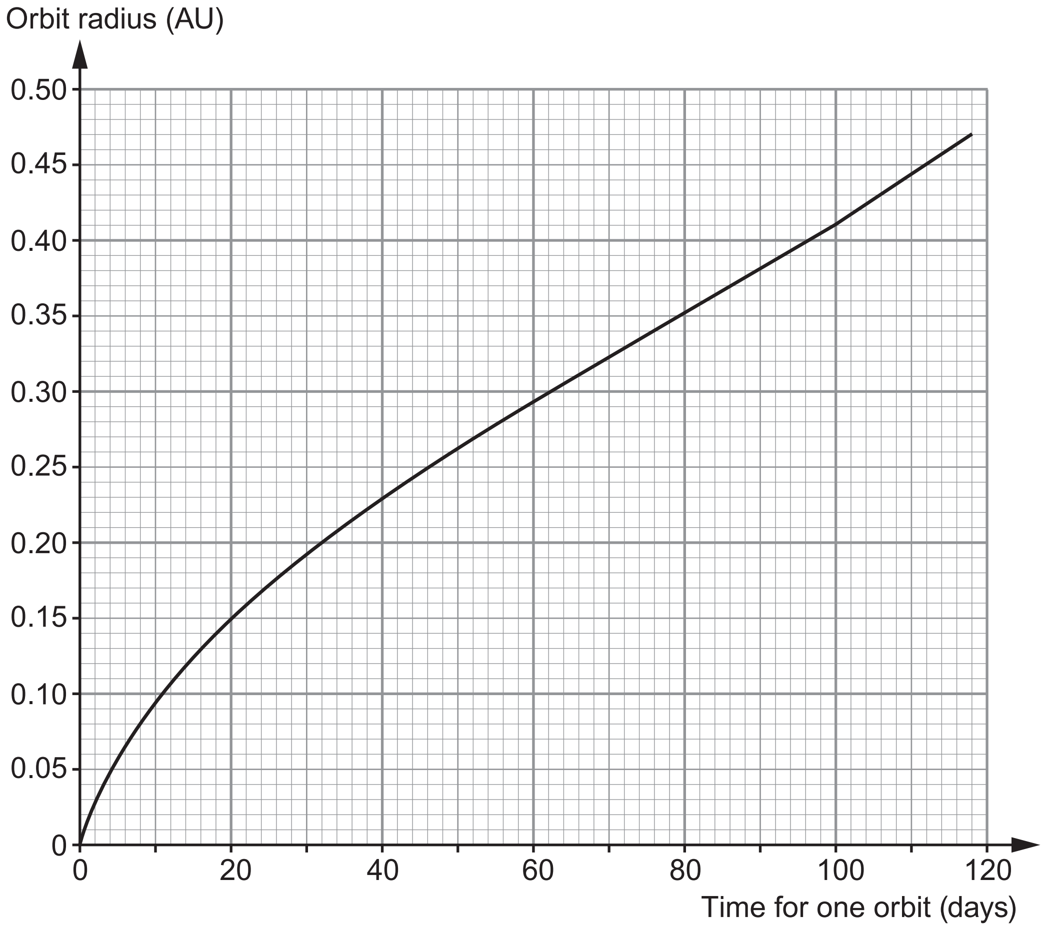
.............................................................................................................................................

.............................................................................................................................................

.............................................................................................................................................

Examiner only

(ii) Some of the data from the table is used to plot a graph.

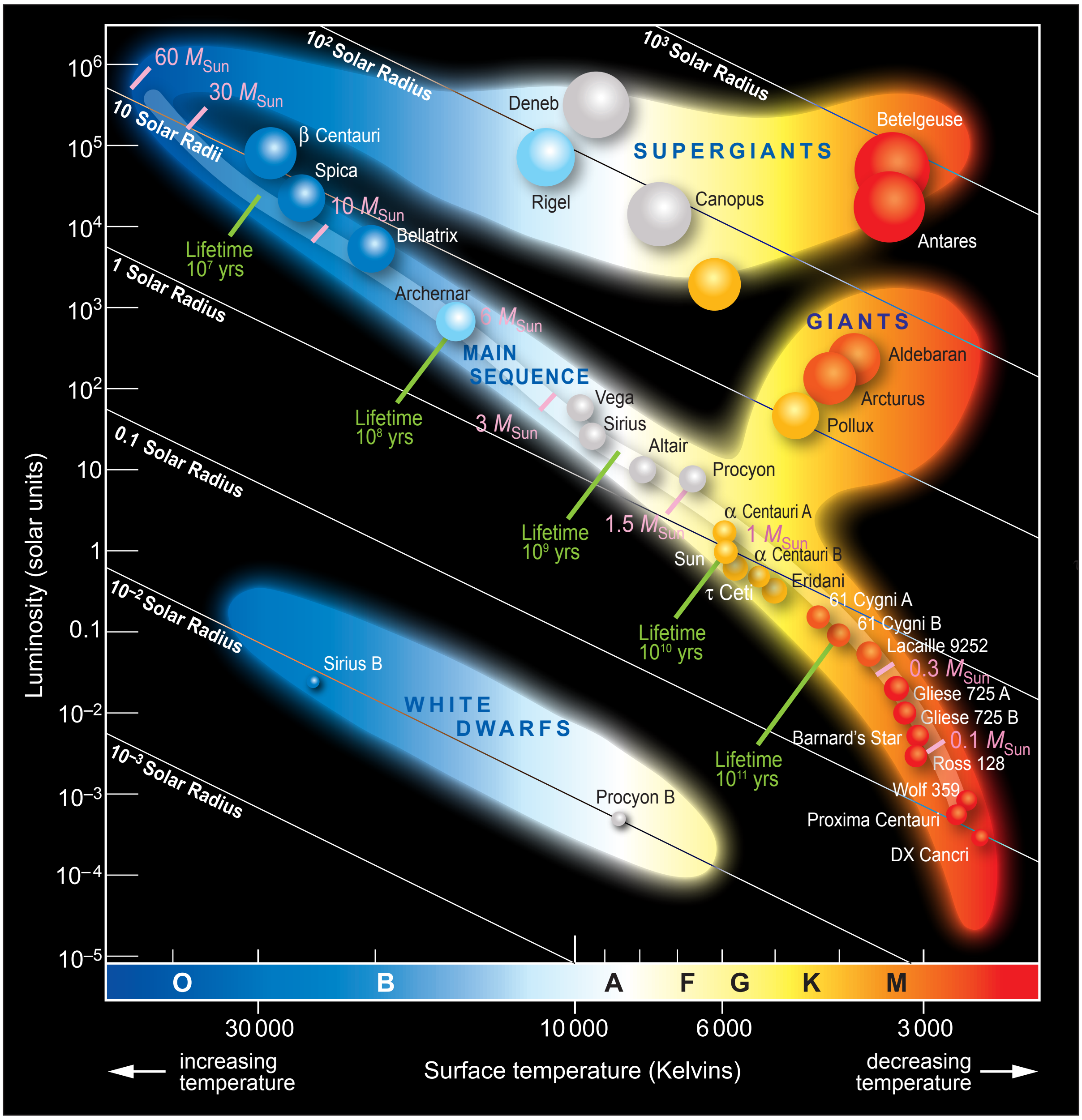



The orbit time of **exoplanet f** was found using the transit method. Its orbit time was measured as 46.7 days. Use the graph to **complete Table 2** with an estimate of a value for the missing orbit radius. [1]

3420UB01
09

13

**5.** The Hertzsprung-Russell (H-R) diagram below is a means of displaying the properties of stars and representing their evolutionary paths.



Examiner only

(a) Tick (✓) the boxes next to the **three** correct statements about the stars shown in the H-R diagram. [3]

The larger the mass of a main sequence star the longer its lifetime. ☐

The largest supergiant is Betelgeuse. ☐

The radius of a white dwarf is approximately 100 times smaller than the radius of the Sun. ☐

The hottest star is β Centauri. ☐

Centauri A is big enough to become a supergiant. ☐

The redder a star the hotter it is. ☐

(b) (i) It is claimed that, as our Sun leaves the main sequence, its surface temperature will increase in the next stage of its life and then decrease as it enters the last stage of its life. Explain whether you agree with this statement. You should include data in your answer. [3]

..........

..........

..........

..........

..........

(ii) Explain, in terms of named forces, the changes our Sun will undergo as it leaves the main sequence until it reaches the end of its life. [3]

..........

..........

..........

..........

..........

Examiner only

*(c)* Explain how the life cycle of stars contributed to the origin of our Solar System. [3]

..............................................................................................................................................................

..............................................................................................................................................................

..............................................................................................................................................................

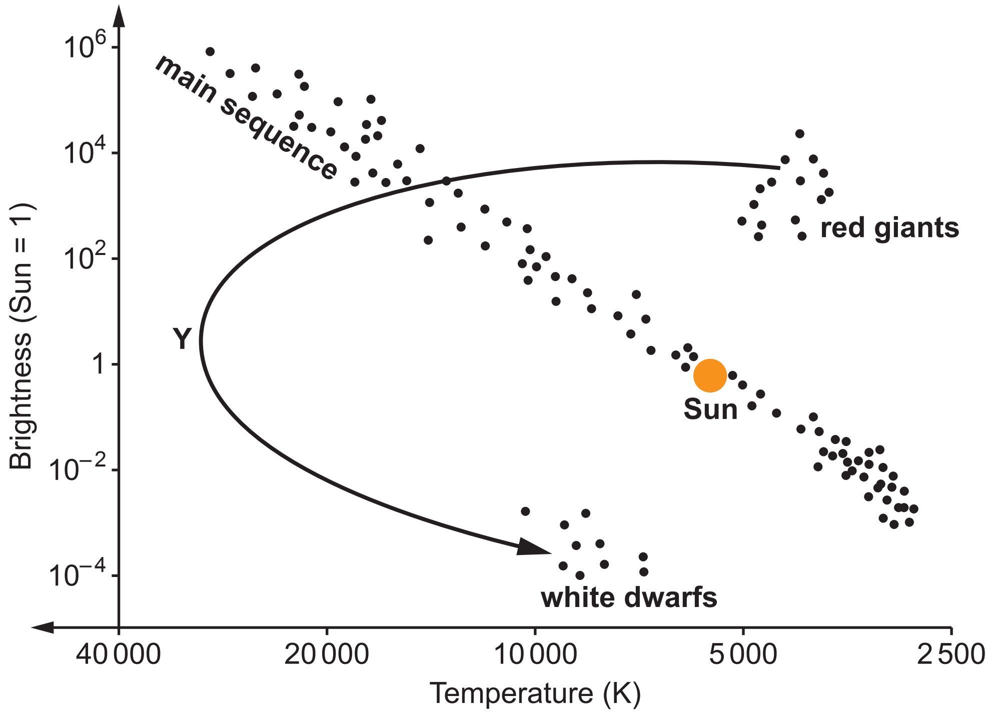
..............................................................................................................................................................

..............................................................................................................................................................

12

Examiner only

**5.** The HR diagram below shows the evolutionary path of our Sun.




(a) Currently the Sun is part of the main sequence but when it evolves into a red giant, its temperature decreases and its brightness increases.

(i) The arrow shows the path the Sun will take as it evolves from a red giant to a white dwarf.
Use information from the HR diagram, to describe how the **temperature** of the Sun changes along this path.
Use the point **Y** in your answer to help with your description. [2]

..............................................................................................................................................

..............................................................................................................................................

..............................................................................................................................................

(ii) Use information from the HR diagram, to describe how the **brightness** of the Sun changes when it evolves from a red giant into a white dwarf. [1]

..............................................................................................................................................

Examiner only

(b) The life cycle of a high mass star is different from that of the Sun. Starting with the main sequence, state, in order, the remaining stages in its life cycle. [3]

..........................................................................................................................................................

..........................................................................................................................................................

..........................................................................................................................................................

(c) An information table obtained from the internet shows some distances used in astronomy.

| Distance | Equivalent |
|---|---|
| 1 light year (l-y) | 63 241 astronomical units (AU) |
| 1 astronomical unit (AU) | $1.5 \times 10^8$ km |

Sirius B is a white dwarf. It is **8.6 light years** away from Earth.
Use information from the table to calculate the distance between Earth and Sirius B.
Give your answer **in metres**. [3]

distance = .................................................... m

9

Examiner only

**2.** The table below gives data about some objects in our solar system.

| Object | Mass (units) | Diameter (km) | Length of day (hours) | Year length (days) | Orbital speed (km/s) | Mean temperature (°C) | Distance from Sun (units) |
|---|---|---|---|---|---|---|---|
| Mercury | 0.330 | 4 879 | 4 222.6 | 88 | 47.4 | 167 | 0.39 |
| Venus | 4.87 | 12 104 | 2 802 | 225 | 35 | 464 | 0.72 |
| Earth | 5.97 | 12 756 | 24 | 365 | 29.8 | 15 | 1.00 |
| Moon | 0.073 | 3 475 | 708.7 | | 1 | –20 | |
| Mars | 0.642 | 6 792 | 24.7 | 687 | 24.1 | –65 | 1.52 |
| Jupiter | 1 898 | 142 984 | 9.9 | 4 331 | 13.1 | –110 | 5.20 |
| Saturn | 568 | 120 536 | 10.7 | 10 747 | 9.7 | –140 | 9.54 |
| Uranus | 86.8 | 51 118 | 17.2 | 30 589 | 6.8 | –195 | 19.18 |
| Neptune | 102 | 49 528 | 16.1 | 59 800 | 5.4 | –200 | 30.06 |
| Pluto | 0.0146 | 2 370 | 163.3 | 90 560 | 4.7 | –225 | 39.53 |

(a) (i) Use the information from the table to **tick** (✓) the **two** correct statements below. [2]

Neptune is hotter than the Moon. ☐

The mean temperature of the Moon is 5 degrees less than the Earth. ☐

A year on Earth is about 4 times longer than a year on Mercury. ☐

Mercury orbits the Sun with a speed around 10 times greater than Pluto. ☐

(ii) Pluto was once considered to be a planet but is now classed as a dwarf planet. Use the data in the table to suggest a reason for the change. [1]

..............................................................................................................................................

..............................................................................................................................................

Examiner only

(iii) Ceres is another dwarf planet and it is the only dwarf planet located in the asteroid belt. Estimate its distance from the Sun. [1]

Distance from the Sun = .................................... units

(b) Elin concludes that rocky planets with the greatest mass have the shortest day length. Explain the extent to which the data supports this conclusion. [2]

..........................................................................................................................................................

..........................................................................................................................................................

..........................................................................................................................................................

6

3420UB01 07

Examiner only

(c) Use an equation from page 2 to calculate the velocity of the skateboarder after he catches the rugby ball. [2]

Velocity = .................................. m/s

(d) Tomos suggests that **kinetic energy** is conserved in this collision.
Explain, by using calculations, whether or not you agree.
Use an equation from page 2. [4]

.............................................................................................................................................

.............................................................................................................................................

12

**END OF PAPER**

Examiner only

**3.** Alpha Centauri is a star of similar size to our Sun.
It is 4.37 light years from Earth.
It is a main sequence star.

(a) State how long light takes to travel from Alpha Centauri to Earth. [1]

time = ..............................................

(b) (i) Explain, in terms of named forces, why Alpha Centauri is currently stable. [2]

..............................................................................................................................................

..............................................................................................................................................

..............................................................................................................................................

(ii) Explain, in terms of named forces, the next stage in the life of Alpha Centauri. [2]

..............................................................................................................................................

..............................................................................................................................................

..............................................................................................................................................

(c) Explain why the mass of hydrogen present in Alpha Centauri will decrease as it gets older. [2]

..............................................................................................................................................

..............................................................................................................................................

..............................................................................................................................................

7

Examiner only

**3.** Rigel is a high-mass blue star, 870 light-years away from Earth.
Rigel is currently on the main sequence.

(a) Use an equation from page 2 to calculate the distance of Rigel from Earth in metres. [3]
Speed of light, $c = 3 \times 10^8$ m/s
1 year = 31 536 000 s

distance = .................................... m

(b) (i) Complete the boxes to show the remaining stages in Rigel's life cycle. [3]

| main sequence star | → | | → | | → | |
|---|---|---|---|---|---|---|

(ii) Explain why large mass stars are important in the creation of new solar systems. [2]

..............................................................................................................................................

..............................................................................................................................................

..............................................................................................................................................

(iii) Describe how a solar system forms. [2]

..............................................................................................................................................

..............................................................................................................................................

..............................................................................................................................................

10